

```

CREATE DATABASE pharmacy_db;
USE pharmacy_db;

CREATE TABLE CUSTOMER (
    CustomerID      INT AUTO_INCREMENT PRIMARY KEY,
    FirstName       VARCHAR(50) NOT NULL,
    LastName        VARCHAR(50) NOT NULL,
    ContactNumber    VARCHAR(20) NOT NULL,
    Address         VARCHAR(100),
    DateOfBirth     DATE
);

CREATE TABLE PHARMACIST (
    PharmacistID    INT AUTO_INCREMENT PRIMARY KEY,
    Name            VARCHAR(50) NOT NULL,
    ContactNumber    VARCHAR(20) NOT NULL,
    Email           VARCHAR(100) UNIQUE,
    Position        VARCHAR(50)
);

CREATE TABLE SUPPLIER (
    SupplierID      INT AUTO_INCREMENT PRIMARY KEY,
    SupplierName     VARCHAR(100) NOT NULL,
    ContactNumber    VARCHAR(20) NOT NULL,
    Address         VARCHAR(100),
    Email           VARCHAR(100) UNIQUE
);

CREATE TABLE MEDICINE (
    MedicineID      INT AUTO_INCREMENT PRIMARY KEY,
    MedicineName     VARCHAR(100) NOT NULL,
    Category        VARCHAR(50),
    Dosage          VARCHAR(50),
    UnitPrice       DECIMAL(10,2) NOT NULL CHECK (UnitPrice >= 0),
    StockQuantity   INT NOT NULL DEFAULT 0 CHECK (StockQuantity >= 0),
    ExpiryDate      DATE NOT NULL,
    SupplierID      INT,
    FOREIGN KEY (SupplierID) REFERENCES SUPPLIER(SupplierID)
        ON UPDATE CASCADE ON DELETE SET NULL
);

CREATE TABLE PRESCRIPTION (
    PrescriptionID   INT AUTO_INCREMENT PRIMARY KEY,
    PrescriptionDate DATE NOT NULL,
    DosageInstructions VARCHAR(255),
    Duration        VARCHAR(50),
    CustomerID       INT NOT NULL,
    PharmacistID     INT NOT NULL,
    FOREIGN KEY (CustomerID) REFERENCES CUSTOMER(CustomerID)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    FOREIGN KEY (PharmacistID) REFERENCES PHARMACIST(PharmacistID)
        ON UPDATE CASCADE ON DELETE RESTRICT
);

CREATE TABLE PRESCRIPTION_ITEM (
    PrescriptionItemID INT AUTO_INCREMENT PRIMARY KEY,
    PrescriptionID     INT NOT NULL,

```

```

MedicineID          INT NOT NULL,
Quantity            INT NOT NULL CHECK (Quantity > 0),
Dosage              VARCHAR(50),
Frequency           VARCHAR(50),
Duration            VARCHAR(50),
FOREIGN KEY (PrescriptionID) REFERENCES PRESCRIPTION(PrescriptionID)
    ON UPDATE CASCADE ON DELETE CASCADE,
FOREIGN KEY (MedicineID) REFERENCES MEDICINE(MedicineID)
    ON UPDATE CASCADE ON DELETE RESTRICT
);

```

```

CREATE TABLE PURCHASE (
    PurchaseID        INT AUTO_INCREMENT PRIMARY KEY,
    PurchaseDate      DATE NOT NULL,
    SupplierID        INT NOT NULL,
    FOREIGN KEY (SupplierID) REFERENCES SUPPLIER(SupplierID)
        ON UPDATE CASCADE ON DELETE RESTRICT
);

```

```

CREATE TABLE PURCHASE_ITEM (
    PurchaseItemID    INT AUTO_INCREMENT PRIMARY KEY,
    PurchaseID        INT NOT NULL,
    MedicineID        INT NOT NULL,
    QuantityPurchased INT NOT NULL CHECK (QuantityPurchased > 0),
    UnitCost          DECIMAL(10,2) NOT NULL CHECK (UnitCost >= 0),
    FOREIGN KEY (PurchaseID) REFERENCES PURCHASE(PurchaseID)
        ON UPDATE CASCADE ON DELETE CASCADE,
    FOREIGN KEY (MedicineID) REFERENCES MEDICINE(MedicineID)
        ON UPDATE CASCADE ON DELETE RESTRICT
);

```

Indexes (beyond the PK/FK indexes MariaDB creates automatically)

```

CREATE INDEX idx_medicine_name ON MEDICINE(MedicineName);
CREATE INDEX idx_medicine_expiry ON MEDICINE(ExpiryDate);
CREATE INDEX idx_prescription_date ON PRESCRIPTION(PrescriptionDate);
CREATE INDEX idx_customer_lastname ON CUSTOMER(LastName);

```